Introduction to Algorithms and Data Structures

week 6: Trees

SAXION UNIVERSITY OF APPLIED SCIENCES

Trees

- **Discuss work last lesson**
- Trees IRL
- Binary trees
- Exercises / Homework

- Where you able to do the exercises?
  1. Yes! All exercises are done!
  2. No.., Because I have these questions
  3. I haven't prioritized this subject enough at the moment to do the assignments

Trees

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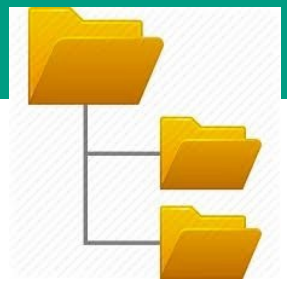
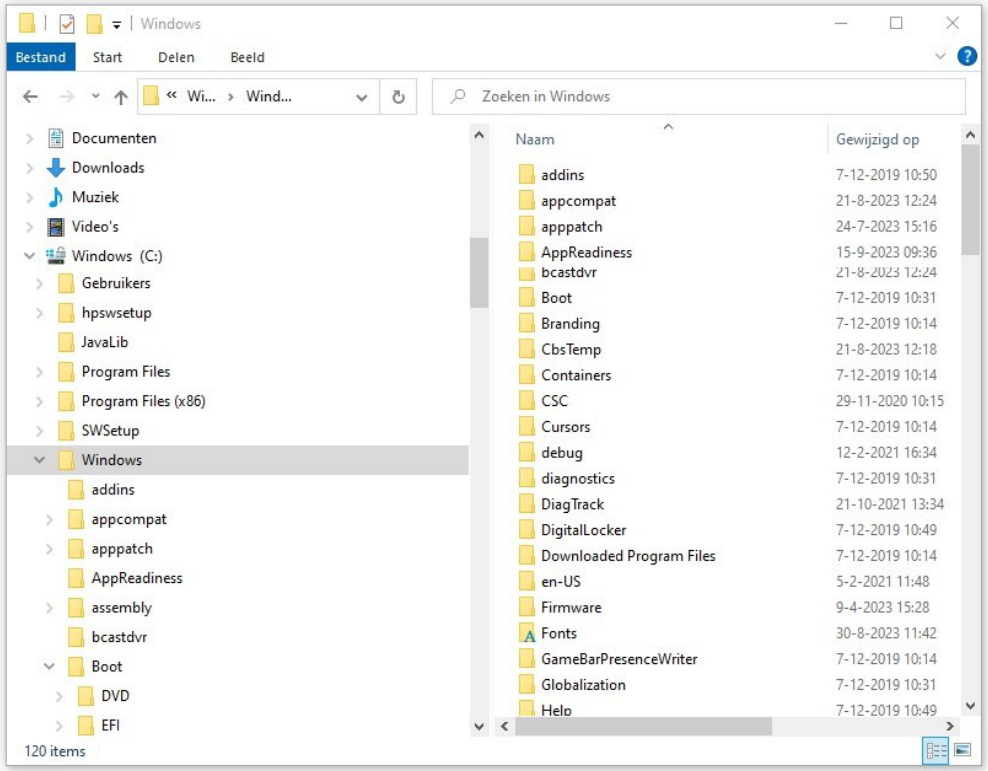
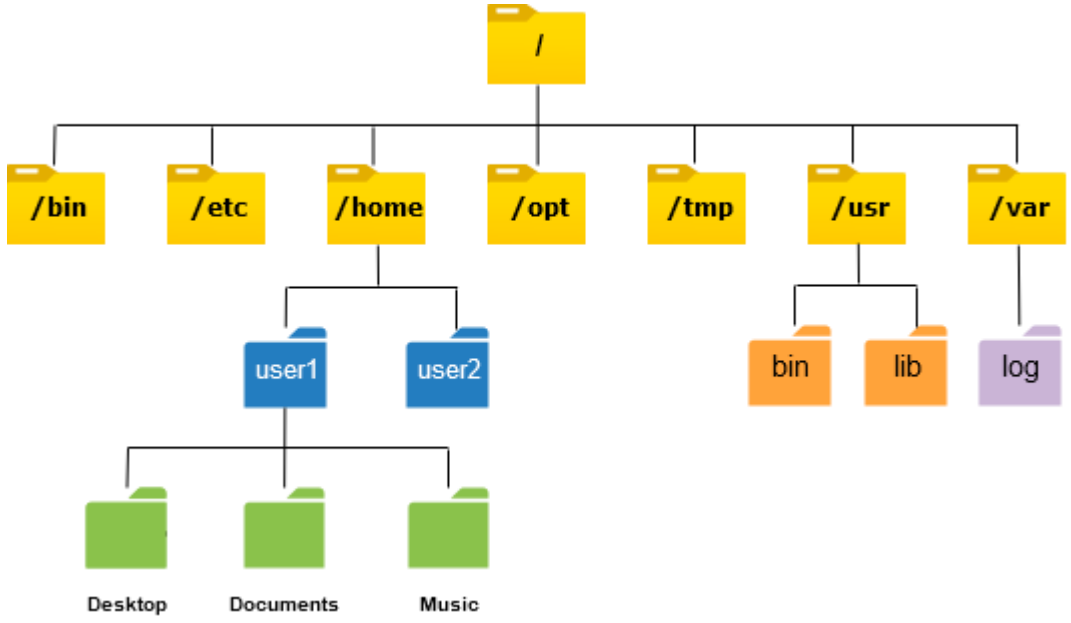


Directories

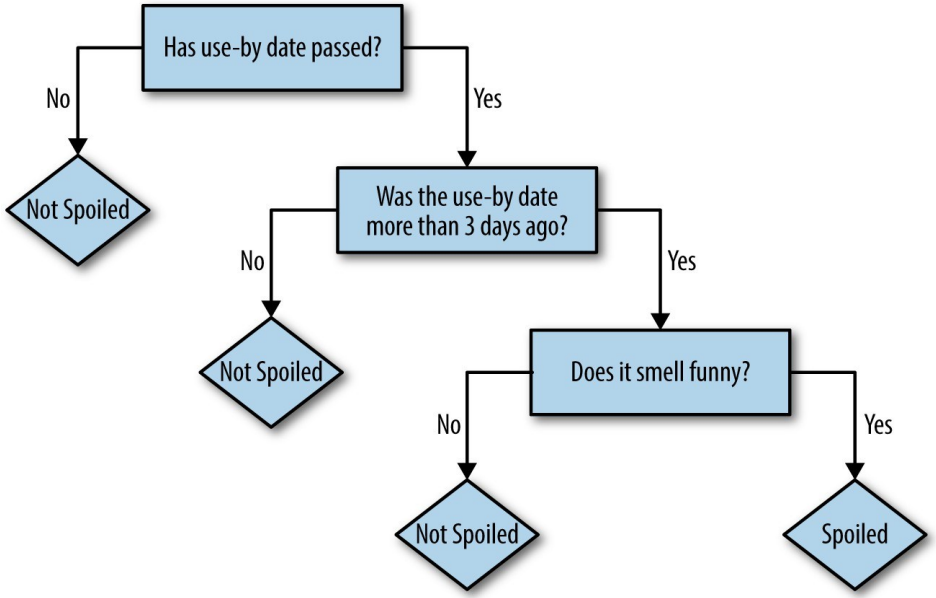
- Directory structure:
  - Most operating systems have a file system with a hierarchical structure (files and subdirectories)





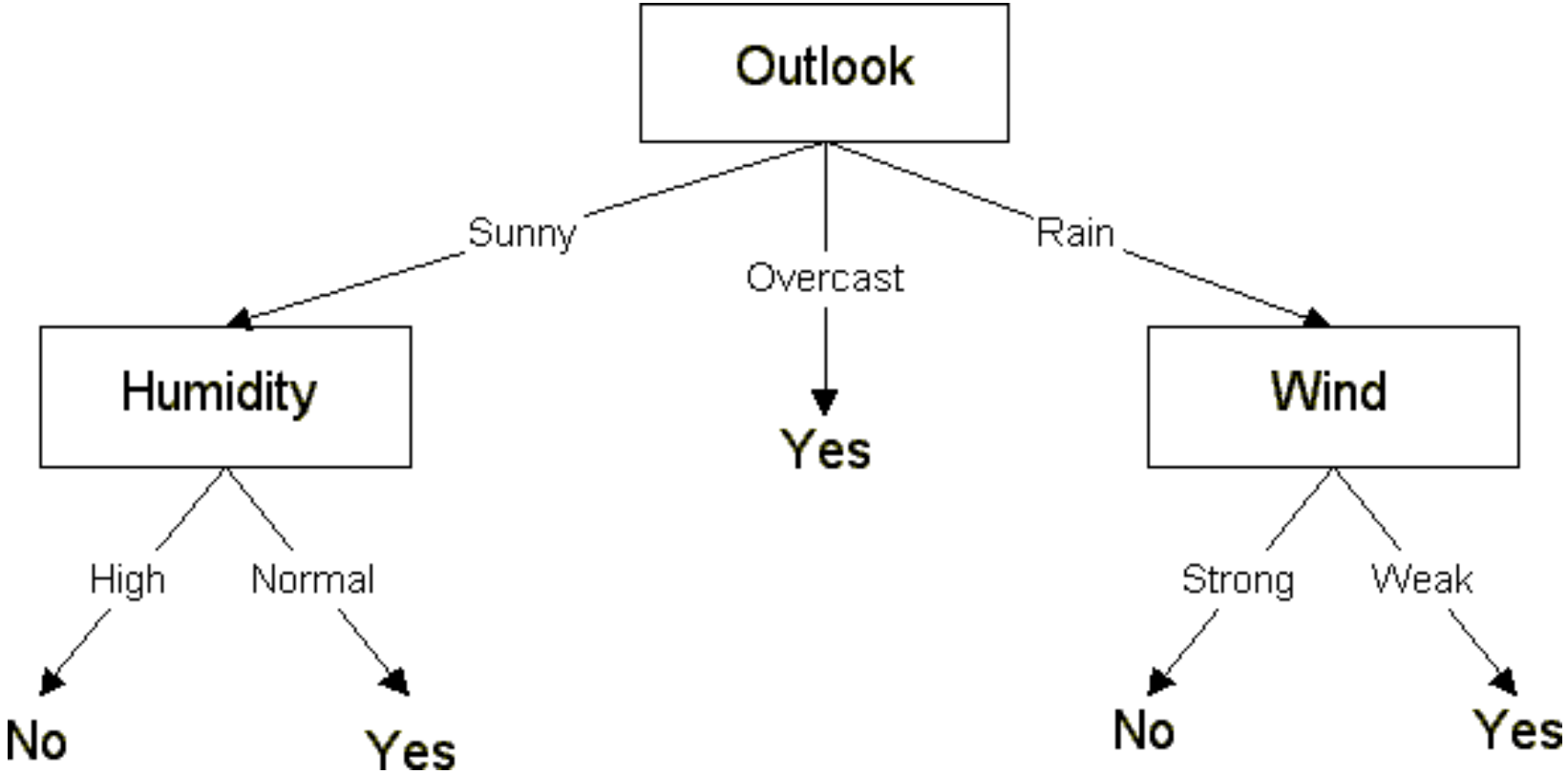
Decision trees

- Decision tree:
  - To find the solution to a problem, just ask the right questions
- Decision tree features:
  - Each internal node contains a question
  - Each branch represents an answer
  - Each leaf represents a decision



Decision trees

- Example:
  - Decision tree "Will you play tennis?"



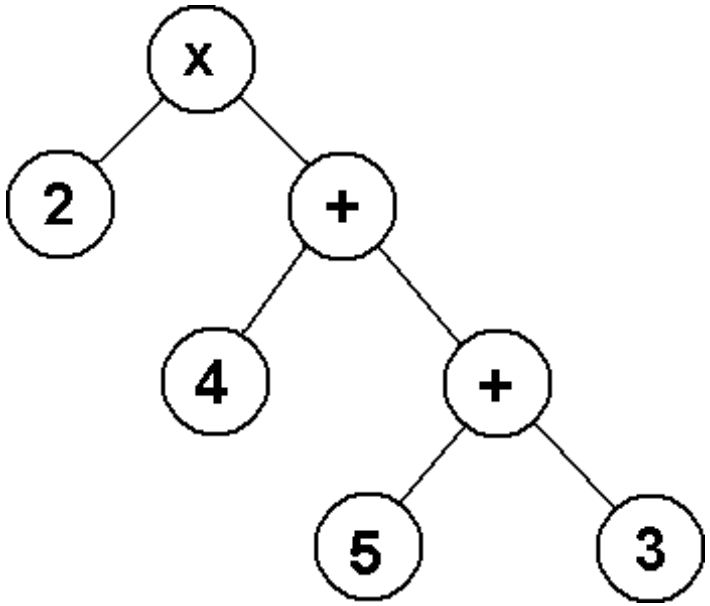
Expression Trees

- Expression tree:
  - Each node contains an operator or a number

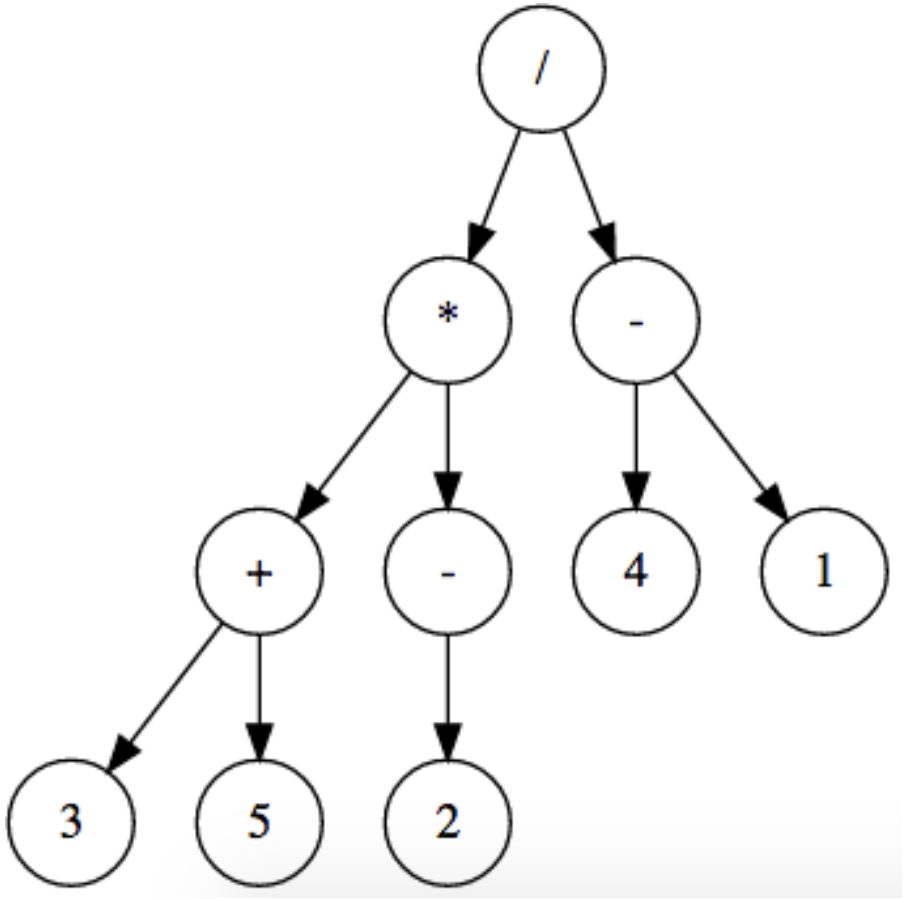
- Expression tree properties:
  - An operator node has one or more subtrees.
  - A constant (number, string etc.) node is a leaf
  - The outcome of an expression tree is:
    - the value of the constant, or
    - the operator applied to the outcomes of the subtrees (recursion)

Expression Trees - example

2 * (4 + (5 + 3))

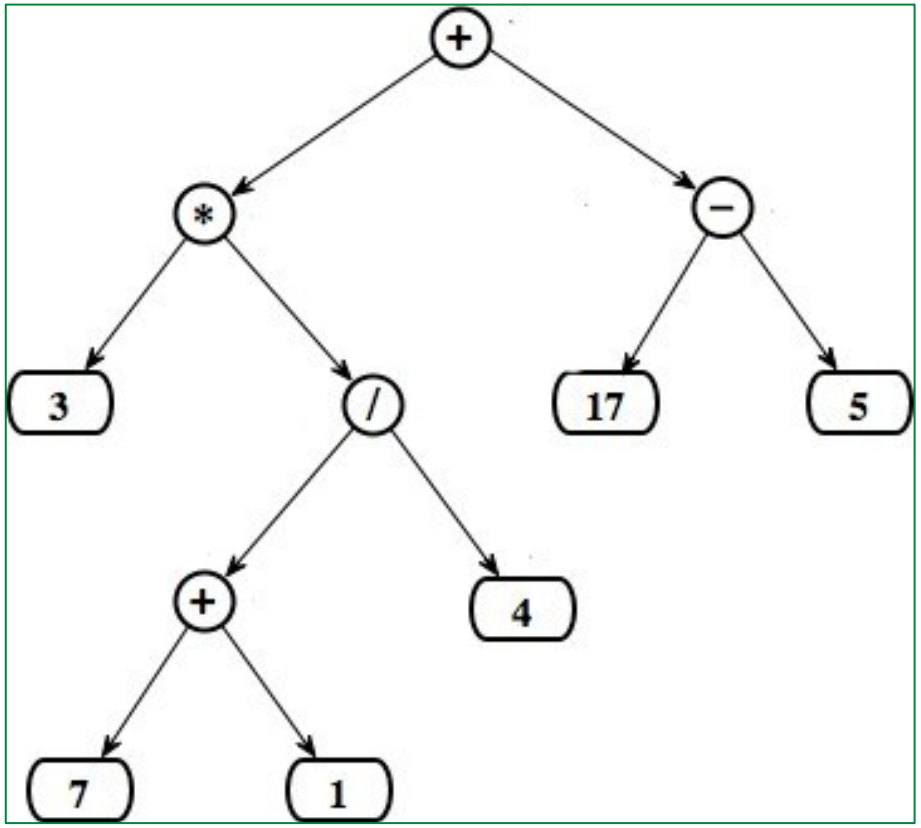


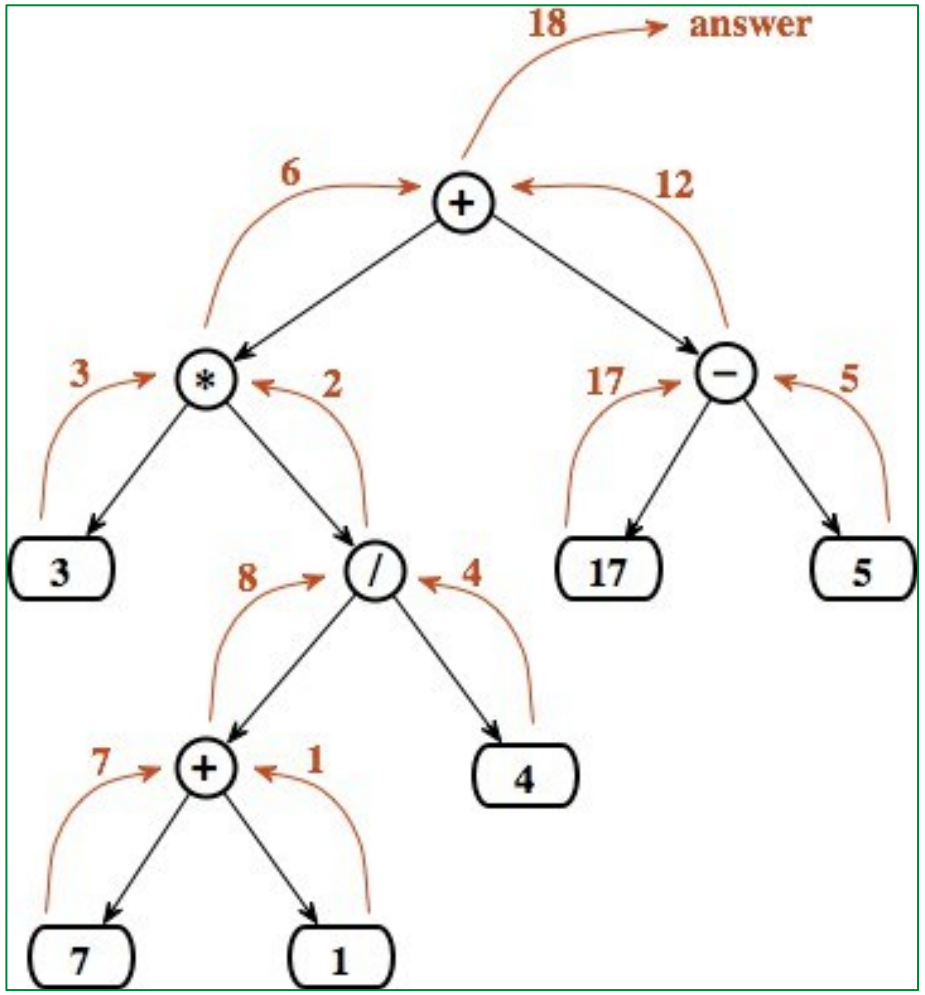
((3 + 5) * (-2)) / (4 - 1)



Expression Trees - calculate

3 * ((7 + 1) / 4) + (17 - 5) = 18





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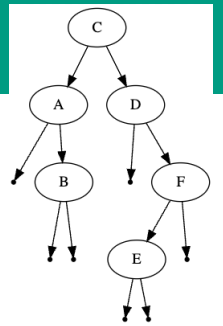
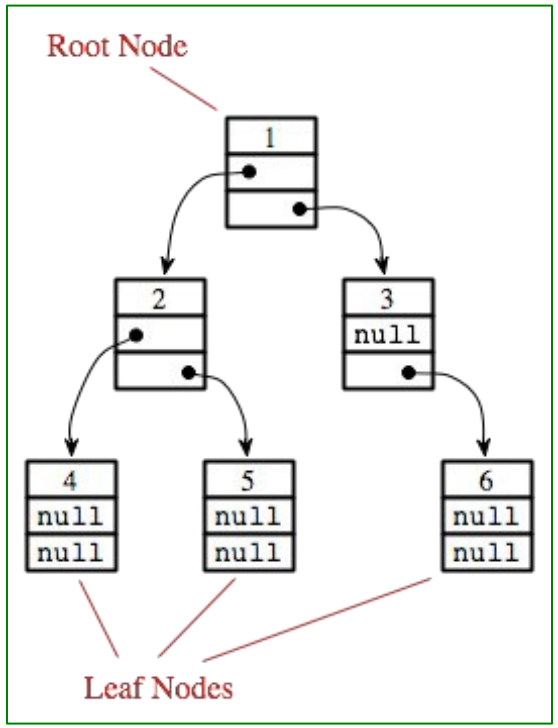
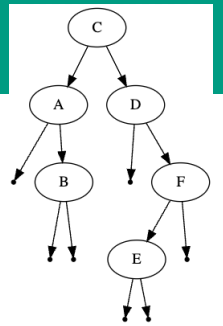


Binary trees

- Binary tree:
  - Sort of singly linked list, where we call the first element “root” and every node has 2 next nodes instead of 1 and every nod is a tree in itself, just like a list node is a list in itself.



- Properties:
  - Each Node contains data and left and right (child) Node.
  - Root = starting node of a tree.
  - Leaf = a node without any child node.



Binary tree traversal

- A Node is a recursive data structure:
  - Left and Right are also nodes.
  - Every node is a tree in itself.
  - Printing all nodes in a tree is therefore a recursive function:
    - Start at the root node (if any).
    - Print the value of the node.
    - Recursively call printing of the left child node (if any – base case)
    - Recursively call printing of the right child node (if any – another base case)

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Now it’s your turn!